

Resolving single-qubit gate amplitude and phase errors in a single experiment

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1 Introduction

Gate errors in superconducting qubit systems can be broadly divided into amplitude errors, described by the Pauli σ_x -type error operators, and phase errors, described by the Pauli σ_z -type error operators. Amplitude errors can be calibrated by scaling the amplitude of the real part of control drives, meanwhile phase errors can be suppressed by adding an imaginary part and/or dynamical detuning, commonly referred to as DRAG correction. Both amplitude scaling and DRAG crucially rely on error characterization to obtain the proper correction: at least two different experiments, in the right order, must be done to determine two mutually dependent parameters. To date, none has demonstrated the ability to learn about the two errors in a single experiment.

We provide here a theoretical framework to learn about the two errors in a single experiment and experimentally demonstrated suppression of both errors in a superconducting qutrit, whose universal control is based on two-level subsystem rotations. Gate performance are then benchmarked against the traditional error suppression procedure, highlighting the efficiency of our method. The improvement is finally quantitatively analyzed through the use of qutrit randomized benchmarking, revealing **question mark**. Our results open up a new direction to efficient error characterization and calibration in a superconducting qubit system, which is becoming more and more vital as the ever-growing size of these quantum information processing units enter the era of practical utility.

Note to self: Find the answer to the question mark. Add numbers. Ramsey error filter is very powerful, the 2008 paper of Lucero [1] is still being referenced in the 2026 paper active leakage cancellation of Google's Yang [2].

2 Results

2.1 Protocol

Starting from the amplified phase error protocol ([3]), hereafter referred as APE, we modify the gate sequence to allow for learning of both phase and amplitude errors. First, we describe the APE sequence. It starts with a prepared superposition of $|1\rangle$ and $|2\rangle$. Next, we apply $(X_{-\pi/2}X_{\pi/2})^{2n+1}$ to the prepared state, where $X_{\pi/2}$ is the gate we wish to characterize. If $X_{\pi/2}$ is limited by phase errors, characterized by an error parameter δ , the sequence $(X_{-\pi/2}X_{\pi/2})^{2n+1}$ approximates a Z rotation of argument $(2n+1)\delta$. Finally, we apply a $\pi/2$ rotation with axis of rotation ϕ varying over the range of $\phi \in [0, 2\pi)$. This converts the relative phase $(2n+1)\delta$ accumulates on the prepared eigenstates into a measurable population. In the original protocol of APE, by measuring the population of $|2\rangle$ (or $|1\rangle$) as a function of ϕ , $p_2(\phi)$ one can learn about phase errors δ in the $X_{\pi/2}$ gate.

We introduce here our modified protocol, termed DRAPE (DRag-APE). In DRAPE, the final rotation is fixed at $\phi = \pi/2$, or a Y rotation. At this optimal ϕ point, population variation with respect to ϕ , that is $\partial p_2 / \partial \phi$, is maximal. We state here one of our main results: an erroneous $X'_{\pi/2}$ can be effectively

described using two error parameters ϵ , representing the overrotated angle (or amplitude error), and δ , representing the aggregate phase errors,

$$R'_x(\pi/2) = R_\delta(\pi/2 + \epsilon)R_z(-2\delta). \quad (1)$$

Enabled by this parametrization, at the end of the DRAPE sequence we obtain the state $|\Psi_{\text{DRAPE}}\rangle$,

$$|\Psi_{\text{DRAPE}}\rangle = \frac{1}{2} \begin{pmatrix} -i(1 - \sin \epsilon)M_{11} + S_k(\theta)(1 + i) \cos^2 \epsilon \sin \delta e^{i\delta} + (1 + \sin \epsilon)e^{i2\delta}M_{22} \\ -e^{i\delta} \cos \epsilon [iM_{11} + e^{i2\delta}M_{22}] + S_k(\theta) \cos \epsilon \sin \delta e^{i2\delta}(1 + i)(\sin \epsilon - i) \end{pmatrix}, \quad (2)$$

where $S_k(\theta) = \frac{\sin(k\theta)}{\sin \theta}$ is related to the Chebyshev polynomials of the second kind and the characteristic angle $\theta = \cos^{-1}[\frac{1}{2}((1 + \sin \epsilon) + (1 - \sin \epsilon) \cos 2\delta)]$, and

$$M_{11} = \frac{S_k(\theta)}{2} [(1 - \sin \epsilon)e^{-i2\delta} + (1 + \sin \epsilon)] - S_{k-1}(\theta), \quad (3)$$

$$M_{22} = M_{11}^*. \quad (4)$$

Since both ϵ and δ are encoded inside $|\Psi_{\text{DRAPE}}\rangle$, this is a classic parameter estimation problem. We can further our understandings of how both ϵ and δ affect the Ramsey fringe in a DRAPE experiment by expanding to first order in δ , while keeping ϵ arbitrary. In this limit,

$$\theta \approx \delta \sqrt{2(1 - \sin \epsilon)}, \quad (5)$$

$$S_k(\theta) \approx k, S_{k-1}(\theta) \approx k - 1, \quad (6)$$

$$M_{11} \approx 1 - i\delta k(1 - \sin \epsilon), M_{22} = M_{11}^*. \quad (7)$$

Substituting these equalities into $|\Psi_{\text{DRAPE}}\rangle$ and discarding $O(\delta^2)$ terms, we find the (already normalized) form of the final state to first order in δ ,

$$|\Psi_{\text{DRAPE}}\rangle \approx |\psi_0\rangle + \delta|\psi_1\rangle, \quad (8)$$

$$|\psi_0\rangle = \frac{1}{2} \begin{pmatrix} (1 + \sin \epsilon) - i(1 - \sin \epsilon) \\ -(1 + i) \cos \epsilon \end{pmatrix}, \quad (9)$$

$$|\psi_1\rangle = \frac{1}{2} \begin{pmatrix} k \cos^2 \epsilon + i(2 + k(1 - \sin \epsilon)^2) \\ \cos \epsilon [k(\sin \epsilon - 1) - i(k \sin \epsilon + k + 2)] \end{pmatrix}. \quad (10)$$

We thus find the first-order expressions for the probabilities of measuring the two states $|m\rangle, |n\rangle$ involved in a DRAPE experiment,

$$P(|m\rangle) = \frac{1 + \sin^2 \epsilon}{2} - \delta \cos^2 \epsilon (k + 1 - 2k \sin \epsilon), \quad (11)$$

$$P(|n\rangle) = \frac{1}{2} \cos^2 \epsilon + \delta \cos^2 \epsilon (k + 1 - 2k \sin \epsilon), \quad (12)$$

where we recognize that

- When $\delta = 0$, we lose the k -dependence and all probabilities converge at $\cos^2(\epsilon)/2$.
- Away from $\delta = 0$, but $\delta \ll 1$ (small), we have the probabilities behaving as straight lines as a function of δ .

The straight lines are described by a slope depending on k and ϵ . We can describe these k -dependent straight lines as having y -intercept of $\cos^2 \epsilon/2$ and slope $\cos^2 \epsilon [k(1 - 2 \sin \epsilon) - 1]$. This confirms the behavior of our dataset. Since the two probabilities sum up to unity, we from now on look into $P(|n\rangle)$.

2.2 Procedure to find DRAG and amplitude correction

Suppose that there exists a DRAG solution β_{DRAPE} such that $\beta = \beta_{\text{DRAPE}}$ corresponds to suppressed phase errors, i.e. $\delta \rightarrow 0$, we can model this by

$$P(|n\rangle) = \frac{1}{2} \cos^2 \epsilon + \xi(\beta - \beta_{\text{DRAPE}}) \cos^2 \epsilon (k + 1 - 2k \sin \epsilon). \quad (13)$$

In practice, finding ϵ and β_{DRAPE} corresponds to finding two values that best describes the dataset. This problem can be solved as follow (in the real paper, this will probably goes into the Supplementary).

- Step 1. We observe that for a fixed k , the population is a simple line as a function of β

$$p(|n\rangle) = \underbrace{\xi \cos^2 \epsilon [k(1 - 2 \sin \epsilon) + 1]}_{M_k} \times \underbrace{\beta + \frac{1}{2} \cos^2 \epsilon - \beta_{\text{DRAPE}} \xi \cos^2 \epsilon [k(1 - 2 \sin \epsilon) + 1]}_{C_k}, \quad (14)$$

$$= M_k \beta + C_k, \quad (15)$$

where M_k is the slope and C_k is the intercept.

- Step 2. Since the slope M_k is a function of ξ and ϵ , we obtain a rough estimate of ϵ by performing a linear fit of M_k as a function of k . The slope is then $S_k = \xi \cos^2 \epsilon (1 - 2 \sin \epsilon)$ and the intercept $B_k = \xi \cos^2 \epsilon$. Thus

$$\epsilon_k = \sin^{-1} \left(\frac{B_k - S_k}{2B_k} \right), \quad (16)$$

$$\xi_k = \frac{B_k}{\cos^2 \epsilon_k}. \quad (17)$$

- Step 3. To obtain β_{DRAPE} , we recognize the fact that it is the slope of C_k as a function of M_k with intercept $(\cos^2 \epsilon_k)/2$. We do another linear regression and obtain β_{DRAPE} .
- Step 4. We use these estimates for inputs of a global nonlinear least squares fit across k and β . The goal is to minimize the total residual across all k and β simultaneously,

$$\min_{\epsilon, \beta_0, \xi} \sum_{k, \beta} [p_2(\beta, k) - p'_2(\beta, k, \epsilon, \beta_0, \xi)]^2, \quad (18)$$

where p_2 is the data we obtained from experiments and p'_2 is Eqn. (13)

3 Execution

Starting from the amplified phase error protocol (Lucero 2008), we modify the gate sequence to allow for learning of both phase and amplitude errors.

References

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